Debate Only When Necessary: Adaptive Multiagent Collaboration for Efficient LLM Reasoning

Sugyeong Eo, Hyeonseok Moon, Evelyn Hayoon Zi, Chanjun Park, Heuiseok Lim*
Korea University
{djtnrud,glee889,evehy,bcj1210,limhseok}@korea.ac.kr

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Abstract

Multiagent collaboration has emerged as a promising framework for enhancing the reasoning capabilities of large language models (LLMs). Despite improvements in reasoning, the approach introduces substantial computational overhead resulting from iterative agent interactions. Furthermore, engaging in unnecessary debates increases the risk of generating erroneous responses. To address these challenges, we propose Debate Only When Necessary (DOWN), an adaptive multiagent debate framework that selectively activates debate based on the confidence score of the agent's initial response. Debate is activated only for queries requiring further deliberation, during which agents refine their outputs by referencing peer responses and associated confidence scores. Evaluations on benchmarks show that DOWN improves efficiency by up to six times while preserving or even outperforming the performance of existing methods. Further analysis indicates that DOWN effectively mitigates the risk of error propagation stemming from the unnecessary debate process. These findings demonstrate the effectiveness of our approach in delivering high-performance LLM solutions at a lower computational cost.

1 Introduction

Building on the remarkable advancements in large language models (LLMs), recent research has increasingly focused on extending their capabilities to address complex real-world problems (Yao et al., 2023; Fan et al., 2024; Chen et al., 2024b). Among various research directions, multiagent collaboration has emerged as a promising approach, inspired by human decision-making processes in complex problem-solving (Minsky, 1988; Li et al., 2023; Chen et al., 2024a; Wang et al., 2025; Wu et al., 2023; Du et al., 2024). By engaging in structured debate, LLM agents systematically exchange perspectives and iteratively cross-examine each other's reasoning to refine their responses. This collaborative process facilitates divergent thinking and enhances the reasoning capabilities of LLMs (Liang et al., 2024; Chen et al., 2024b; Chan et al., 2024).

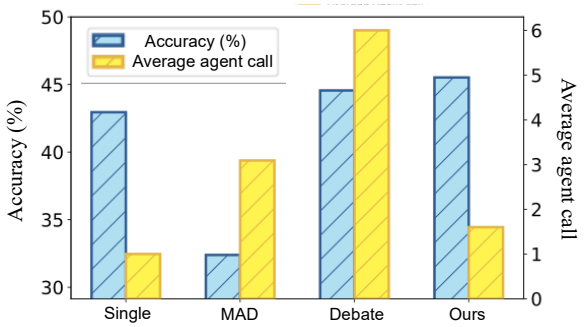

Figure 1: Comparison of accuracy and average agent calls across various multiagent debate methods

Despite these advantages, multiagent collaboration systems exhibit several key limitations. From an efficiency perspective, iterative interactions among agents inherently require multiple agent calls, leading to increased latency and higher inference costs (Snell et al., 2024; Kapoor et al., 2024). Meanwhile, when agents engage in redundant or unnecessary debate, there is a higher likelihood of generating errors that may subsequently be propagated by other agents in the system (Wang et al., 2024). Figure 1 illustrates both of these issues by plotting the accuracy and average agent calls of different multiagent debate methods. In the Debate system, additional debate rounds improve performance while incurring a sixfold increase in computational overhead. In the MAD framework, despite using more agent calls than the single agent baseline, its accuracy paradoxically declines. Regarding the practical application of multiagent collaboration systems, these challenges emphasize the need for an optimized collaboration approach (Kapoor

*Corresponding Author

et al., 2024; Tran et al., 2025).

To address these limitations, we propose Debate Only When Necessary (DOWN), an adaptive multiagent collaboration framework that identifies queries requiring debate and selectively facilitates the debate process accordingly. This framework employs the confidence score to quantify the internal certainty of LLM outputs, a measure that has been extensively utilized to enhance model performance and trustworthiness (Razghandi et al., 2025; Taubenfeld et al., 2025; Chen et al., 2024a). In the DOWN framework, the initial agent response is generated as the first step. A confidence score is computed during response generation and used to determine whether debate should be initiated. If the confidence score exceeds a threshold, the debate is skipped, and we regard the initial response as a final answer. If further refinement is required, agents engage in debate to enhance response accuracy. When the collaboration begins, a confidence-guided debate is conducted, during which agents refine their responses by considering both the responses of other participating agents and their confidence scores. This approach encourages the refinement of responses by utilizing the most persuasive aspects of agent responses.

Our experiments employ models of varying scales, including approximately 8B and 70B parameter models, as well as GPT-4o-mini, evaluated on the MUSR (Sprague et al., 2024) and StrategyQA (Geva et al., 2021) benchmarks. The results demonstrate that adaptive debate invoking significantly reduces computational overhead while maintaining or even surpassing the performance of full-debate baselines. Notably, we reveal that this conditional debate serves as a safeguard against cascading errors, effectively enhancing the advantages of agent collaboration. Our contribution is threefold:

- We propose Debate Only When Necessary (DOWN), an adaptive multiagent framework that selectively initiates debate based on the initial response. To the best of our knowledge, this is the first study to explore conditional debate in multiagent systems grounded in initial responses, maximizing efficiency while preserving the benefits of discussion.
- Extensive experiments across diverse model sizes and configurations establish the effectiveness of the mechanism. We further find that adaptively engaging debate contributes to mitigating error propagation.
- We establish that the confidence-guided debate process enables the selective integration of reliable responses, emphasizing the effectiveness of multiagent collaboration.

2 Related Work

LLM-based Multiagent Collaboration Drawing inspiration from human collaborative problem-solving behavior, multiagent collaboration systems leverage collective intelligence to improve decision-making. Studies have demonstrated that LLM-powered multiagent systems promote divergent thinking (Xiong et al., 2023; Liu et al., 2024; Liang et al., 2024) and improve reasoning capabilities (Li et al., 2023; Yin et al., 2023; Zhuge et al., 2023). With these advantages, multiagent collaboration is leveraged for diverse NLP applications: mitigating hallucinations (Fang et al., 2025), aggregating knowledge across multiple specialized LLMs (Wang et al., 2025), generating novel scientific ideas and insights (Su et al., 2024), evaluating LLM-generated responses (Chan et al., 2024), and refining datasets for instruction fine-tuning (Li et al., 2024a). These advancements highlight the growing impact of multiagent collaboration.

Debate Structures in Multiagent Collaboration Systems Recent studies have developed debate structures to optimize the benefits of multiagent collaboration. For instance, Du et al. (2024) introduces a framework in which agents iteratively refine their responses based on peer-generated outputs. Liang et al. (2024) propose a structured debate format that assigns distinct roles to encourage divergent thinking. Wang et al. (2024) develop a discussion system in which agents are organized into multiple groups to engage in discussions. However, iterative debate frameworks face a fundamental trade-off between efficiency and accuracy (Kapoor et al., 2024; Snell et al., 2024). To this end, Li et al. (2024b) restrict discussions to local neighbors, while Zhou et al. (2025) introduce a shortcut mechanism that shares similarities with our approach. However, both methods still incur non-trivial agent calls. The former requires neighbor interactions, and the latter relies on self-consistency, with both involving at least five agent calls per sample.

Additionally, iterative debates may propagate errors or introduce a trade-off between diversity and quality (Wang et al., 2024; Kapoor et al., 2024; Li et al., 2025). Our approach focuses on these limitations by proposing a resource-efficient and

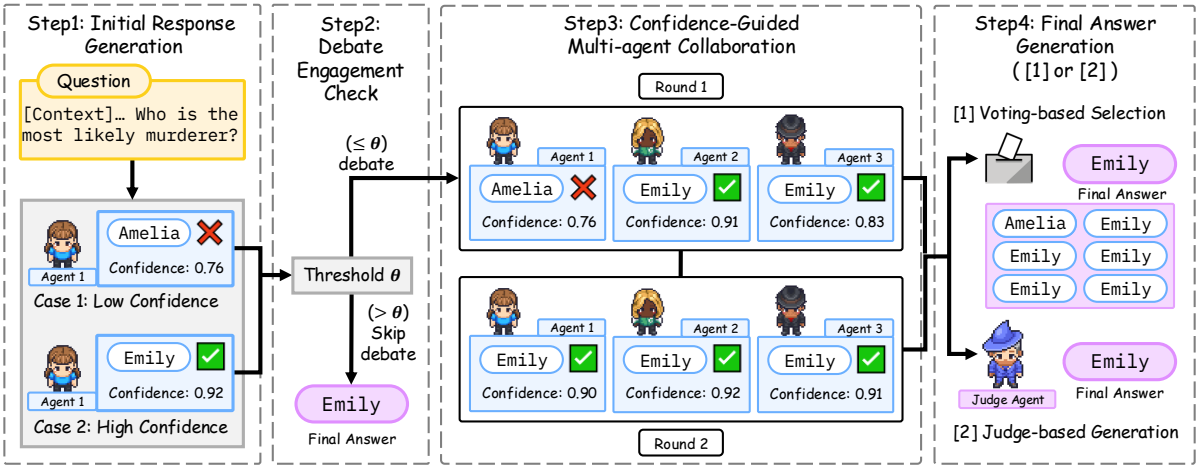

Figure 2: Overview of the Debate Only When Necessary (DOWN) framework. DOWN consists of four stages: (1) the initial agent generates a response, during which the model's confidence score is extracted. (2) if the confidence score exceeds a threshold value, the response is accepted without debate to improve efficiency, otherwise a multiagent debate is activated. (3) agents refine their responses by referencing peer outputs and associated confidence scores. (4) the final answer is selected via majority voting or designated judge agent.

performance-effective debate framework.

3 Debate Only When Necessary

The collaboration framework consists of four steps: (1) initial response generation, (2) debate engagement check, (3) confidence-guided multiagent collaboration, and (4) final response generation. The confidence score derived from the model during initial response generation determines its subsequent progression. If the score exceeds a predefined threshold, the further debate process is skipped. Otherwise, we proceed with the debate, allowing agents to engage in confidence-guided debate with other agents. We illustrate the overall multiagent collaboration framework in Figure 2.

3.1 System Pipeline

Step 1: Initial Response Generation The model's confidence score obtained during initial response generation determines the progression of the collaboration process. Based on its importance, we adopt two strategies for multiagent configurations: (1) A homogeneous-agent configuration where all collaborating agents use the same model (e.g., Llama) and (2) a mixed-agent configuration where the initial agent is randomly selected for each query (e.g., Llama, Qwen, GPT-4o-mini).

For a given query q, an initial agent $\mathcal{A}_1$ from the set of agents $\{\mathcal{A}\}_{i=1}^{N}$ generates an initial response r_1 with the reason for the answer. During this process, a confidence score c_1 is extracted to quantify the model's certainty. We first obtain the token logit $L(t_i)$, where $t_i \in r_1$, from the hidden representation of the generated response. These logits are then passed through a softmax function to yield the probability distribution $P(t_i)$ over tokens. To obtain a robust estimate of confidence, we apply normalization by averaging the token probabilities over the generated response. The confidence score c_1 is defined as:

$$c_1 = \frac{1}{|r_1|} \sum_{i=1}^{|r_1|} P(t_i), \quad \text{where} P(t_i) = \frac{e^{L(t_i)}}{\sum_j e^{L(t_j)}}. \tag{1}$$

For models that cannot access internal logits, we utilize verbalized confidence as an alternative method.

Step 2: Debate Engagement Check Along with the initial response, we extract a confidence score c_1 reflecting the model's certainty in its answer. The confidence score is compared against a threshold score θ:

- (1) High confidence ($> \theta$): The response is accepted as the final response without further collaboration, optimizing efficiency by avoiding unnecessary computations.
- (2) Low confidence ($\leq \theta$): The initial response is deemed unreliable, activating the collaboration process to improve response quality.

In this context, the threshold is a hyperparameter, where a higher threshold places more emphasis